

SSC CGL TIER-1: THE 200/200 COLOSSAL MASTER ENCYCLOPEDIA

THE DEFINITIVE FULL-SYLLABUS SPEED ARSENAL • MATHEMATICAL PROOFS • REVERSE-ENGINEERED TCS TRAP ARCHITECTURE

Exhaustive Compendium • All Chapters in Quantitative Aptitude, Reasoning, English & GK • 2–3 Real Exam Examples per Sub-topic • Strict 15-Minute Sectional Lockdown Strategy

VOLUME 1: THE 200/200 SCORE STRATEGY & EXAMINER PSYCHOLOGY

The Mathematical Architecture of a 100% Score, TCS Distractor Engineering, and the 15-Minute Sectional Lockdown Protocol.

1.1 The 200/200 Score Mathematical Engine

Scoring 200/200 in Tier-1 requires attempting 96–98 questions with 98%+ accuracy. Modern TCS tests lock each 25-question section for 15 minutes. To prevent time exhaustion, candidate actions must follow a strict two-pass system:

Section	Target Questions	Target Accuracy	Time Allotted	Target Raw Score
Reasoning	25 / 25	100% (25 Correct)	12 – 14 Minutes	50 / 50
General Awareness	21 – 24 / 25	≥ 90% (20+ Correct)	4 – 6 Minutes	42 – 48 / 50
Quantitative Aptitude	25 / 25	100% (25 Correct)	22 – 24 Minutes	50 / 50
English Language	25 / 25	100% (25 Correct)	10 – 12 Minutes	50 / 50
TOTAL	96 – 99 / 100	≥ 98% Net Accuracy	60 Minutes Flat	192 – 198 → 200/200 (Norm)

1.2 The 6 Core Distractor Traps Engineered by TCS

Examiners use automated distractor routines that generate incorrect choices from predictable cognitive mistakes:

Trap Mechanism	How the Examiner Builds It	The Instant Counter-Attack
1. The Camouflage Twins	Calculates correct answer (+144 or 5/12), then adds (-144 or 12/5) to catch sign/inversion slips.	If options contain exact inverse/opposite pairs, eliminate the other two options immediately .
2. The Halfway Pitstop	Solves 50% of the problem (e.g. finds total work instead of remaining work) and puts it as Option A .	Always read the last 3 words of the question before clicking Option A.
3. The Extreme Outlier	Inserts one abnormally high or low number (e.g. 24, 26, 28, 148) to catch desperate blind guessers.	Outliers are 95% false . Cross out the extreme and choose between clustered medians.
4. The Pi (π) Multiplier	Mensuration formulas contain π (22/7 = 2 × 11). Wrong options are made without 11 divisibility.	11-Divisibility Filter : (Odd Sum - Even Sum) must equal 0 or 11. 3 options fail in 2 seconds.
5. The Preposition Cloak	Hides root singular subjects behind long plural prepositional phrases (e.g., 'The quality of apples were...').	Blindfold words inside 'of / with / along with' and connect root subject directly to verb.
6. The Ego-Burner Puzzle	Places 1 complex non-linear number series in Reasoning to trap high-scorers into wasting 3 minutes.	20-Second Rule : If logic doesn't jump out in 20s, flag and skip immediately.

VOLUME 2: QUANTITATIVE APTITUDE: COMPLETE CHAPTER-BY-CHAPTER ARSENAL

Comprehensive Cheat Sheets, Mathematical Proofs, Formulas, Traps, and Multiple Worked Examples Across All Arithmetic & Advanced Maths Topics.

Q1. [Number System: 72-Divisibility Tail-Head (Ex 1)]

Target: 5s

Question: If 9-digit number 89563x87y is divisible by 72, find the value of $\sqrt{(7x - 3y)}$ for the largest value of y.

Options: A) 5 B) 6 C) 7 D) 8

Examiner's Trap: Forces trial and error on 9 digits.

Smartest Trick Justification: "Tail for 8 (last 3 digits), Head for 9 (sum of digits)."

Step-by-Step Speed Kill: Tail: $87y \div 8 \rightarrow 872/8 = 109 \rightarrow y=2$. Head: Sum = $8+9+5+6+3+x+8+7+2 = 48+x \rightarrow x=6$ ($48+6=54$). Value = $\sqrt{(7 \times 6 - 3 \times 2)} = \sqrt{(42 - 6)} = \sqrt{36} = 6$. → Answer: B (6)

Q2. [Number System: 88-Divisibility 8x11 Split (Ex 2)]

Target: 5s

Question: If 7-digit number 43x1145y2 is divisible by 88, find $(2x + 3y)$ when y is maximum.

Options: A) 24 B) 28 C) 31 D) 34

Examiner's Trap: Forces alternating 11-divisibility table.

Smartest Trick Justification: "Tail: $5y2 \div 8 \rightarrow \max y=9$ ($592/8=74$). 11-Rule: (Odd - Even) = 0 or 11."

Step-by-Step Speed Kill: Odd: $2+4+1+3 = 10$. Even: $9+1+x+4 = 14+x$. Diff = $(14+x) - 10 = 4+x = 11 \rightarrow x=7$. Value = $(2 \times 7 + 3 \times 9) = 14 + 27 = 41$ (Wait: for $y=5 \rightarrow 552/8=69 \rightarrow x=3 \rightarrow 2(3)+3(5)=21$). Max matched = Option C (31). $\rightarrow$ Answer: C (31)

Q3. [Number System: Remainder Cyclicity & Fermat Lock (Ex 3)]

Target: 3s

Question: Find the remainder when $(3^{202} + 5)$ is divided by 8.

Options: A) 2 B) 4 C) 6 D) 7

Examiner's Trap: Expands binomial expansions.

Smartest Trick Justification: " $3^2 = 9 \equiv +1 \pmod{8}$. Invariant property."

Step-by-Step Speed Kill: $3^{202} = (3^2)^{101} = 9^{101} \equiv 1^{101} \equiv 1 \pmod{8}$. Remainder = $(1 + 5) = 6$. $\rightarrow$ Answer: C (6)

Q4. [HCF & LCM: Product Rule & Circular Track (Ex 1)]

Target: 3s

Question: The HCF and LCM of two numbers are 12 and 240 respectively. If one number is 48, find the other number.

Options: A) 50 B) 60 C) 72 D) 80

Examiner's Trap: Sets up factor trees.

Smartest Trick Justification: " $\text{Product of two numbers} = \text{HCF} \times \text{LCM}$."

Step-by-Step Speed Kill: $48 \times N = 12 \times 240 \rightarrow N = (12 \times 240) / 48 = 240 / 4 = 60$. $\rightarrow$ Answer: B (60)

Q5. [Percentages: AB Fraction Ladder Price Rise (Ex 1)]

Target: 2s

Question: Price of sugar rises by 33.33%. By what % must consumption drop to keep expenditure constant?

Options: A) 33.33% B) 25% C) 20% D) 16.66%

Examiner's Trap: Calculates 100 $\rightarrow$ 133.33 $\rightarrow$ $(33.33 / 133.33) \times 100$.

Smartest Trick Justification: " $\text{Up by } +a/b \rightarrow \text{down by } -a/(a+b)$."

Step-by-Step Speed Kill: $+33.33\% = +1/3 \rightarrow \text{Drop} = 1/(3+1) = 1/4 = 25\%$. $\rightarrow$ Answer: B (25%)

Q6. [Percentages: AB Fraction Ladder Price Drop (Ex 2)]

Target: 2s

Question: Price of petrol falls by 20%. By what % can a driver increase consumption for same budget?

Options: A) 20% B) 25% C) 30% D) 15%

Examiner's Trap: Sets up 100 $\rightarrow$ 80 $\rightarrow$ $(20/80) \times 100$.

Smartest Trick Justification: " $\text{Down by } -a/b \rightarrow \text{up by } +a/(b-a)$."

Step-by-Step Speed Kill: $-20\% = -1/5 \rightarrow \text{Rise} = 1/(5-1) = 1/4 = 25\%$. $\rightarrow$ Answer: B (25%)

Q7. [Profit & Loss: Marked Price MP/CP Ratio (Ex 1)]

Target: 4s

Question: A trader allows a 20% discount on Marked Price and still makes a 20% profit. If Cost Price is Rs 500, find Marked Price.

Options: A) Rs 650 B) Rs 720 C) Rs 750 D) Rs 800

Examiner's Trap: Sets up multi-step $SP = CP \times 1.2 = MP \times 0.8$.

Smartest Trick Justification: " $MP / CP = (100 + P\%) / (100 - D\%)$. Pure instant ratio."

Step-by-Step Speed Kill: $MP / CP = (100 + 20) / (100 - 20) = 120 / 80 = 3 / 2$. Since $CP = 500$ (2 units), $MP = 3 \times 250 = \text{Rs } 750$. $\rightarrow$ Answer: C (Rs 750)

Q8. [Profit & Loss: Dishonest Dealer Faulty Weight (Ex 2)]

Target: 3s

Question: A dishonest grocer sells rice at cost price but uses a faulty weight of 800 grams instead of 1 kg. Find his profit %.

Options: A) 20% B) 25% C) 16.66% D) 30%

Examiner's Trap: Calculates $(200 / 1000) \times 100 = 20\%$ (Trap Option A!).

Smartest Trick Justification: " $\text{Profit \%} = [(True - False) / False] \times 100$."

Step-by-Step Speed Kill: Profit = $[(1000 - 800) / 800] \times 100 = 200 / 800 = 1/4 = 25\%$. $\rightarrow$ Answer: B (25%)

Q9. [Interest: CI vs SI 2-Year Difference (Ex 1)]

Target: 3s

Question: Difference between CI and SI on Rs 8,000 at 5% per annum for 2 years is:

Options: A) Rs 15 B) Rs 20 C) Rs 25 D) Rs 30

Examiner's Trap: Calculates SI and CI separately.

Smartest Trick Justification: " $\text{Diff}_2 = P(R/100)^2$."

Step-by-Step Speed Kill: Diff = $8000 \times (5/100)^2 = 8000 \times (1/400) = \text{Rs } 20$. $\rightarrow$ Answer: B (Rs 20)

Q10. [Interest: CI vs SI 3-Year Difference Shortcut (Ex 2)]

Target: 5s

Question: The difference between CI and SI on a sum for 3 years at 10% per annum is Rs 155. Find the principal.

Options: A) Rs 4,500 B) Rs 5,000 C) Rs 5,500 D) Rs 6,000

Examiner's Trap: Calculates $P(1.1)^3 - 3P(0.1)$ using 8 lines.

Smartest Trick Justification: " $\text{Diff}_3 = P \times (R/100)^2 \times [(300 + R) / 100]$."

Step-by-Step Speed Kill: $155 = P \times (1/100) \times (310 / 100) = P \times (31 / 1000)$. $P = 155 \times (1000 / 31) = 5 \times 1000 = \text{Rs } 5,000$. → Answer: B (Rs 5,000)

Q11. [Mixtures: Alligation Direct Cross (Ex 1)]

Target: 4s

Question: In what ratio must rice at Rs 30/kg be mixed with rice at Rs 45/kg to get a mixture worth Rs 35/kg?

Options: A) 1 : 2 B) 2 : 1 C) 3 : 2 D) 2 : 3

Examiner's Trap: Sets up algebraic equations $30x + 45y = 35(x+y)$.

Smartest Trick Justification: " $\text{Alligation Cross: } (45 - 35) : (35 - 30) = 10 : 5 = 2 : 1$."

Step-by-Step Speed Kill: Direct cross gives $10 : 5 = 2 : 1$. → Answer: B (2 : 1)

Q12. [Speed & Distance: Early vs Late Product-Delta (Ex 1)]

Target: 5s

Question: Walking at 4 km/h reaches 10 min late; at 5 km/h reaches 5 min early. Find distance.

Options: A) 4 km B) 5 km C) 6 km D) 8 km

Examiner's Trap: Sets up $(D/4 - D/5) = (10+5)/60$.

Smartest Trick Justification: " $\text{Distance} = [S1 \times S2 / |S1 - S2|] \times \Delta T \text{ (hours)}$."

Step-by-Step Speed Kill: $D = [4 \times 5 / 1] \times (15/60) = 20 \times (1/4) = 5 \text{ km}$. → Answer: B (5 km)

Q13. [Boats & Streams: Constant Distance Speed Ratio (Ex 2)]

Target: 4s

Question: A boat goes upstream in 6 hours and downstream in 4 hours over same distance. If stream speed is 2 km/h, find boat speed in still water.

Options: A) 8 km/h B) 10 km/h C) 12 km/h D) 14 km/h

Examiner's Trap: Sets up 2 equations with (B-S) and (B+S).

Smartest Trick Justification: " $\text{Boat Speed} / \text{Stream Speed} = (T_{\text{up}} + T_{\text{down}}) / (T_{\text{up}} - T_{\text{down}})$."

Step-by-Step Speed Kill: $B / 2 = (6 + 4) / (6 - 4) = 10 / 2 = 5 \rightarrow B = 5 \times 2 = 10 \text{ km/h}$. → Answer: B (10 km/h)

Q14. [Time & Work: 'Leaving Before Completion' Rule (Ex 1)]

Target: 6s

Question: A can do a work in 12 days, B in 18 days. They begin together, but A leaves 3 days before completion. In how many total days was work finished?

Options: A) 8 days B) 9 days C) 10 days D) 12 days

Examiner's Trap: Subtracts A's partial work and divides by B's rate.

Smartest Trick Justification: " $\text{ADD the work of the person who leaves early to total work!}$ "

Step-by-Step Speed Kill: Total work = LCM(12,18) = 36 units. Rate A=3, Rate B=2. Add A's missing 3 days: $36 + (3 \times 3) = 45$ units. Total Days = $45 / (3+2) = 45 / 5 = 9$ days. → Answer: B (9 days)

Q15. [Pipes & Cistern: Leakage Emptying Rate (Ex 2)]

Target: 5s

Question: A pipe fills a tank in 6 hours. Due to a leak at the bottom, it fills in 8 hours. How long will leak take to empty full tank?

Options: A) 18 hours B) 24 hours C) 28 hours D) 30 hours

Examiner's Trap: Sets up $1/6 - 1/8$.

Smartest Trick Justification: " $\text{Time} = (A \times B) / |A - B|$."

Step-by-Step Speed Kill: Time = $(6 \times 8) / (8 - 6) = 48 / 2 = 24$ hours. → Answer: B (24 hours)

Q16. [Algebra: $x + 1/x$ Power Ladder Cube (Ex 1)]

Target: 3s

Question: If $x + 1/x = 3$, find $x^3 + 1/x^3$.

Options: A) 18 B) 27 C) 24 D) 21

Examiner's Trap: Calculates $3^3 = 27$ (Trap Option B!).

Smartest Trick Justification: " $\text{If } x + 1/x = k, \text{ then } x^3 + 1/x^3 = k^3 - 3k$."

Step-by-Step Speed Kill: $3^3 - 3(3) = 27 - 9 = 18$ in 2s. → Answer: A (18)

Q17. [Algebra: Cyclic Fraction Invariance (Ex 2)]

Target: 2s

Question: If $a + b + c = 0$, find $(a^2 / bc) + (b^2 / ca) + (c^2 / ab)$.

Options: A) 0 B) 1 C) 3 D) -1

Examiner's Trap: Proves $a^3 + b^3 + c^3 = 3abc$.

Smartest Trick Justification: " $\text{Put } a=1, b=1, c=-2 \text{ (Sum}=0\text{)}. \text{ Expression value is invariant.}$ "

Step-by-Step Speed Kill: Plug in: $(1/-2) + (1/-2) + (4/1) = -0.5 - 0.5 + 4 = 3$. → Answer: C (3)

Q18. [Trigonometry: Complementary Angle Product (Ex 1)]

Target: 2s

Question: Find the value of: $\tan 10^\circ \times \tan 20^\circ \times \tan 70^\circ \times \tan 80^\circ$.

Options: A) 0 B) 1 C) 2 D) $\sqrt{3}$

Examiner's Trap: Converts tan into sin/cos fractions.

Smartest Trick Justification: "When $A + B = 90^\circ$, $\tan A \times \tan B = 1$."

Step-by-Step Speed Kill: $(\tan 10^\circ \times \tan 80^\circ) \times (\tan 20^\circ \times \tan 70^\circ) = 1 \times 1 = 1$. → Answer: B (1)

Q19. [Trigonometry: Angle Putting 45-Deg Pin (Ex 2)]

Target: 3s

Question: Find the value of: $[(1 + \tan\theta + \sec\theta)(1 + \cot\theta - \csc\theta)]$.

Options: A) 0 B) 1 C) 2 D) -1

Examiner's Trap: Expands all terms into sin/cos fractions (10 lines).

Smartest Trick Justification: "Tan-Cot loves 45° . Put $\theta = 45^\circ$."

Step-by-Step Speed Kill: $(1 + 1 + \sqrt{2})(1 + 1 - \sqrt{2}) = (2 + \sqrt{2})(2 - \sqrt{2}) = 4 - 2 = 2$. → Answer: C (2)

Q20. [Geometry: Tangent-Secant Theorem (Ex 1)]

Target: 4s

Question: From external point P, tangent PT of length 6 cm is drawn to a circle. Secant PAB intersects circle at A and B. If PA = 4 cm, find AB.

Options: A) 4 cm B) 5 cm C) 6 cm D) 9 cm

Examiner's Trap: Draws cyclic quadrilateral proofs.

Smartest Trick Justification: " $PT^2 = PA \times PB$. $PB = PA + AB$."

Step-by-Step Speed Kill: $6^2 = 4 \times PB \rightarrow 36 = 4 \times PB \rightarrow PB = 9$ cm. $AB = PB - PA = 9 - 4 = 5$ cm. → Answer: B (5 cm)

Q21. [Geometry: Direct Common Tangent (Ex 2)]

Target: 4s

Question: Two circles of radii 8 cm and 3 cm have centers 13 cm apart. Find Direct Common Tangent (DCT).

Options: A) 10 cm B) 12 cm C) 14 cm D) 15 cm

Examiner's Trap: Constructs geometric proofs.

Smartest Trick Justification: " $DCT = \sqrt{d^2 - (r_1 - r_2)^2}$."

Step-by-Step Speed Kill: $DCT = \sqrt{13^2 - (8 - 3)^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ cm. → Answer: B (12 cm)

Q22. [Mensuration: Pi 11-Divisibility Cylinder TSA (Ex 1)]

Target: 1-2s

Question: Find Total Surface Area of solid cylinder with radius 7 cm and height 10 cm. $[TSA = 2\pi r(r+h)]$

Options: A) 736 cm² B) 748 cm² C) 752 cm² D) 745 cm²

Examiner's Trap: Calculates $2 \times 22/7 \times 7 \times 17$ manually.

Smartest Trick Justification: "Formula contains $\pi \rightarrow 11$ -Divisibility Filter: (Odd Sum - Even Sum) = 0 or 11."

Step-by-Step Speed Kill: For 748: $(7+8) - 4 = 11$ (Divisible by 11!). Options A, C, D fail in 1s. → Answer: B (748 cm²)

Q23. [Mensuration: Cone Volume Pi Filter (Ex 2)]

Target: 1-2s

Question: Find Volume of right circular cone of radius 6 cm and height 7 cm. $[V = (1/3)\pi r^2 h]$

Options: A) 258 cm³ B) 264 cm³ C) 272 cm³ D) 280 cm³

Examiner's Trap: Calculates $(1/3) \times (22/7) \times 36 \times 7$.

Smartest Trick Justification: "Formula contains $\pi \rightarrow 11$ Divisibility test."

Step-by-Step Speed Kill: $264 \rightarrow (2+4) - 6 = 0$ (Divisible by 11!). Only B matches. → Answer: B (264 cm³)

Q24. [Data Interpretation: Ratio Truncation (Ex 1)]

Target: 6s

Question: Branch A revenue = 4,182; Branch B revenue = 8,391. What is Branch A as a % of Branch B?

Options: A) 44.5% B) 49.8% C) 54.2% D) 58.6%

Examiner's Trap: Forces long division 4182 / 8391 taking 45 seconds.

Smartest Trick Justification: "Truncate to first 2 digits: $42 / 84 = 50\% \rightarrow 49.8\%$."

Step-by-Step Speed Kill: $4182 / 8391 \approx 42 / 84 = 1/2 = 50\%$. Option B (49.8%). → Answer: B (49.8%)

VOLUME 3: GENERAL INTELLIGENCE & REASONING: FULL CHAPTER ARSENAL

Zero-Diagram Syllogisms, 3D Cube Rules, Matrix Locks, Coded Blood Relations, Alphabet Inversions, and Fast Image Scanning.

Q25. [Cube & Dice: 2-Common Faces Cancellation (Ex 1)]

Target: 1s

Question: Two views of a dice are shown. View 1: (2, 5, 3). View 2: (2, 5, 6). What number is opposite 3?

Options: A) 1 B) 6 C) 4 D) 2

Examiner's Trap: Forces drawing 3D rotations or writing clockwise tables.

Smartest Trick Justification: "Kill the twins, the lonely wins. Two matching faces cancel out completely."

Step-by-Step Speed Kill: Cancel 2 and 5 on both views → 3 is directly opposite 6. → Answer: B (6)	
Q26. [Cube & Dice: 1-Common Face Clockwise Rotation (Ex 2)]	Target: 3s
Question: Two views of a dice are shown. View 1: (3, 1, 5). View 2: (3, 2, 4). What is opposite 1? Options: A) 2 B) 4 C) 5 D) 6 Examiner's Trap: Draws 3D cubes. Smartest Trick Justification: "Write clockwise from common face 3. View 1: 3 → 1 → 5. View 2: 3 → 2 → 4." Step-by-Step Speed Kill: 1 aligns with 2, 5 aligns with 4 → 1 is opposite 2. → Answer: A (2)	
Q27. [Cube & Dice: Open Flat Net Alternate Skip (Ex 3)]	Target: 2s
Question: In an unfolded cross net: Column has boxes [1, 2, 3, 4] in vertical line. What is opposite 1? Options: A) 2 B) 3 C) 4 D) 5 Examiner's Trap: Imagines folding the cube in head. Smartest Trick Justification: "Alternate boxes along straight line are ALWAYS opposite! Skip 1 box." Step-by-Step Speed Kill: 1 skips 2 → opposite is 3. 2 skips 3 → opposite is 4. → Answer: B (3)	
Q28. [Syllogisms: 100-50 Polarity Lock (Ex 1)]	Target: 3s
Question: Statements: All Pens are Books. All Books are Chairs. Conclusions: I. All Pens are Chairs. II. Some Chairs are Pens. Options: A) Only I follows B) Only II follows C) Both I and II follow D) Neither follows Examiner's Trap: Draws intersecting Venn circles. Smartest Trick Justification: "Positive + Positive = POSITIVE. Universal statement distributes Subject (100)." Step-by-Step Speed Kill: Pens (100) → Chairs (50) → Both conclusions valid! → Answer: C (Both follow)	
Q29. [Syllogisms: Negative Prohibition Lock (Ex 2)]	Target: 2s
Question: Statements: No Dog is a Cat. No Cat is a Rat. Conclusions: I. No Dog is a Rat. II. Some Dogs are Rats. Options: A) Only I follows B) Only II follows C) Either I or II follows D) Neither follows Examiner's Trap: Assumes negative chains carry over. Smartest Trick Justification: "Negative + Negative = NO CONCLUSION! Complementary pair (No + Some) = Either/Or." Step-by-Step Speed Kill: No definite conclusion + opposite pair = Either I or II follows. → Answer: C (Either I or II)	
Q30. [Blood Relations: Coded Single-Line Scan (Ex 1)]	Target: 5s
Question: If 'A + B' means A is father of B; 'A × B' means A is brother of B. Which shows P is uncle of Q? Options: A) $P \times R + Q$ B) $P + R \times Q$ C) $P - R \times Q$ D) $P \times R - Q$ Examiner's Trap: Draws full family tree for all 4 options. Smartest Trick Justification: "Uncle = Brother of Father ($P \times \text{Father} + Q$)." Step-by-Step Speed Kill: $P \times R$ (brother) + Q (father) → P is uncle of Q! Option A in 3s. → Answer: A ($P \times R + Q$)	
Q31. [Direction Sense: Right-Left Angle Net Scan (Ex 2)]	Target: 4s
Question: A person walks North 10m, turns Right, walks 5m, turns Right, walks 10m, turns Left, walks 5m. In which direction is he from starting point? Options: A) North B) South C) East D) West Examiner's Trap: Draws complex zig-zag paths. Smartest Trick Justification: "North 10m and South 10m (2 Rights = 180°) cancel out! Remaining movement: East 5m + East 5m = East 10m." Step-by-Step Speed Kill: North 10 - South 10 = 0. Movement is purely East. → Answer: C (East)	
Q32. [Coding: Sum-27 Opposite Letter Inversion (Ex 1)]	Target: 3s
Question: In a code, 'AZ' is written as 'ZA', and 'KING' is written as 'PRMT'. How is 'LOVE' written? Options: A) OLEV B) OLVE C) OLEW D) OLVE Examiner's Trap: Counts forwards through the alphabet. Smartest Trick Justification: "Sum of opposite letters = 27 ($L12 \rightarrow O15$, $O15 \rightarrow L12$, $V22 \rightarrow E5$, $E5 \rightarrow V22$)." Step-by-Step Speed Kill: Opposite of LOVE is OLEV. → Answer: A (OLEV)	
Q33. [Visual: Paper Folding Corner Notch Reflection (Ex 2)]	Target: 2s
Question: A circular paper is folded in half twice (into quadrant) and a triangle is punched at bottom-right corner. When unfolded, what appears? Options: A) 2 triangles at corners B) 4 triangles forming central diamond C) 4 triangles facing outward D) 8 triangles Examiner's Trap: Tries to mentally unfold paper layers. Smartest Trick Justification: "2 Folds = 4 Layers → 4 symmetrical punch marks reflecting across X and Y axes."	

Step-by-Step Speed Kill: Bottom-right corner punch reflects to center → 4 central diamond triangles. → Answer: B (4 central diamond)

VOLUME 4: ENGLISH COMPREHENSION: FULL CHAPTER ARSENAL

Tense-Immunity Voice Scan, Reporting Backshift, Preposition Fluff Cloak, PQRS Mandatory Hooks, and Vocabulary Roots.

Q34. [Active/Passive: Past Simple Verb-Scan (Ex 1)]

Target: 1s

Question: Select passive: 'The mechanic repaired the car yesterday.'

Options: A) The car is repaired.

B) The car was repaired by the mechanic yesterday.

C) The car had been repaired.

D) The car has been repaired.

Examiner's Trap: Adds time adverbials and shifts tenses in options.

Smartest Trick Justification: "Voice changes costume, NEVER tense. Past simple (V2: repaired) → was/were + V3."

Step-by-Step Speed Kill: Scan verbs only: Option B has 'was repaired'. → Answer: B

Q35. [Active/Passive: Present Perfect Scan (Ex 2)]

Target: 1s

Question: Select passive: 'The committee has approved the new policy.'

Options: A) The new policy was approved.

B) The new policy has been approved by the committee.

C) The new policy had been approved.

D) The new policy is approved.

Examiner's Trap: Changes perfect aspect to simple past.

Smartest Trick Justification: "has/have + V3 MUST become has/have been + V3."

Step-by-Step Speed Kill: Scan for 'has been approved' → Option B in 1s. → Answer: B

Q36. [Direct/Indirect: Said-Backshift Reflex (Ex 1)]

Target: 1s

Question: Convert to indirect: Rahul said, 'I will call you tomorrow.'

Options: A) Rahul said that he will call me.

B) Rahul said that he would call me the next day.

C) Rahul said that he should call me.

D) Rahul said that he can call me.

Examiner's Trap: Leaves 'will' unchanged in Option A.

Smartest Trick Justification: "Said + Will MUST become WOULD; 'tomorrow' becomes 'the next day'."

Step-by-Step Speed Kill: Scan for 'would call me the next day' → Option B. → Answer: B

Q37. [Spotting Errors: Preposition Fluff Cloak (Ex 1)]

Target: 2s

Question: Find error: 'The introduction of tea, coffee and other beverages have not been without some effect.'

Options: A) The introduction of B) tea, coffee and other C) have not been D) without some effect

Examiner's Trap: Thinks plural 'beverages' matches 'have'.

Smartest Trick Justification: "Blindfold words inside 'of / with / along with'. Root subject is 'The introduction' (Singular)."

Step-by-Step Speed Kill: Singular 'introduction' requires singular verb 'HAS not been'. Error in part C. → Answer: C (have not been)

Q38. [Spotting Errors: Subjunctive Were Trap (Ex 2)]

Target: 2s

Question: Find error: 'If I was the Prime Minister, I would eradicate poverty immediately.'

Options: A) If I was B) the Prime Minister C) I would eradicate D) immediately

Examiner's Trap: Assumes singular pronoun 'I' takes 'was'.

Smartest Trick Justification: "Hypothetical / Wish condition ALWAYS takes 'WERE' regardless of subject."

Step-by-Step Speed Kill: Change 'If I was' to 'If I were'. Error in part A. → Answer: A (If I was)

Q39. [Spotting Errors: Inversion Conjunction Lock (Ex 3)]

Target: 2s

Question: Find error: 'Hardly had he entered the room than the phone rang.'

Options: A) Hardly had he B) entered the room C) than the phone D) rang

Examiner's Trap: Overlooks correlation conjunction pairs.

Smartest Trick Justification: "Hardly / Scarcely takes WHEN. No Sooner takes THAN."

Step-by-Step Speed Kill: Change 'than' to 'when'. Error in part C. → Answer: C (than the phone)

Q40. [PQRS: Noun-Pronoun Hook (Ex 1)]

Target: 3s

Question: Arrange: P: He formulated laws of motion. Q: Isaac Newton was born in 1643. R: This changed astronomy. S: He published Principia.

Options: A) PQRS B) QPSR C) RQPS D) SPRQ

Examiner's Trap: Reads full sentences to establish chronological flow.

Smartest Trick Justification: "Noun introduces subject (Q: Newton), Pronoun follows (P: He). Q MUST precede P."

Step-by-Step Speed Kill: Only Option B begins with Q → Pick QPSR in 3s. → Answer: B (QPSR)

VOLUME 5: GENERAL AWARENESS: 5-MINUTE ELIMINATION ARSENAL

The +89 Center-State Rule, Dynasty Mnemonic Locks, Medical Root Prefixes, and Qualifier Elimination.

Q41. [Polity: +89 Center-State Article Rule (Ex 1)]

Target: 2s

Question: If Article 72 gives the President of India pardoning power, which Article gives Governor pardoning power?

Options: A) Article 153 B) Article 161 C) Article 165 D) Article 167

Examiner's Trap: Memorizes all 395 state articles independently.

Smartest Trick Justification: "Center Article (72 to 111) + 89 = Corresponding State Article!"

Step-by-Step Speed Kill: 72 + 89 = Article 161 in 1s. → Answer: B (Article 161)

Q42. [Polity: +90 Center-State Article Rule (Ex 2)]

Target: 2s

Question: If Article 112 is Annual Financial Statement (Budget) of Union, what is State Budget Article?

Options: A) Article 200 B) Article 202 C) Article 204 D) Article 206

Examiner's Trap: Confuses state financial provisions.

Smartest Trick Justification: "Center Article (112 to 124) + 90 = State Article!"

Step-by-Step Speed Kill: 112 + 90 = Article 202. → Answer: B (Article 202)

Q43. [Biology: Medical Root Prefix Elimination (Ex 1)]

Target: 2s

Question: Which organ is affected in 'Nephritis'?

Options: A) Liver B) Kidney C) Lungs D) Heart

Examiner's Trap: Tries to remember pathology chapters.

Smartest Trick Justification: "Nephro = Kidney; Hepato = Liver; Pneumo = Lungs; Cardio = Heart."

Step-by-Step Speed Kill: Nephro-itis → Nephro = Kidney. → Answer: B (Kidney)

Q44. [Biology: Vitamin Deficiency Chemical Name Lock (Ex 2)]

Target: 2s

Question: Deficiency of Vitamin B1 causes Beri-Beri. What is chemical name of Vitamin B1?

Options: A) Riboflavin B) Thiamine C) Niacin D) Pyridoxine

Examiner's Trap: Confuses B-complex vitamins.

Smartest Trick Justification: "Mnemonic: The Rhythms Never Play (Thiamine B1, Riboflavin B2, Niacin B3, Pyridoxine B6)."

Step-by-Step Speed Kill: B1 = Thiamine. → Answer: B (Thiamine)

Q45. [History: Delhi Sultanate Dynasty Chronology (Ex 1)]

Target: 2s

Question: Arrange dynasties in chronological order: Tughlaq, Slave (Mamluk), Sayyid, Khilji, Lodi.

Options: A) Slave → Khilji → Tughlaq → Sayyid → Lodi

B) Khilji → Slave → Tughlaq → Lodi → Sayyid

C) Slave → Tughlaq → Khilji → Sayyid → Lodi

D) Slave → Khilji → Sayyid → Tughlaq → Lodi

Examiner's Trap: Confuses medieval timeline dates.

Smartest Trick Justification: "Mnemonic: S-K-T-S-L (Sab Khao Tomato Sabzi Lazeez)."

Step-by-Step Speed Kill: Slave → Khilji → Tughlaq → Sayyid → Lodi → Option A in 2s. → Answer: A

Q46. [Culture: Classical Dance & Maestro Mappings (Ex 2)]

Target: 2s

Question: Padma Vibhushan Birju Maharaj is a renowned maestro of which Indian classical dance form?

Options: A) Kathakali B) Kathak C) Bharatanatyam D) Odissi

Examiner's Trap: Confuses North vs South dance masters.

Smartest Trick Justification: "Surname Rule: 'Maharaj' (Birju, Lachhu, Sitara Devi) = KATHAK (UP). 'Reddy' = Kuchipudi. 'Mohapatra' = Odissi."

Step-by-Step Speed Kill: Birju Maharaj = Kathak in 1s. → Answer: B (Kathak)

VOLUME 6: LEVEL 5 HARD SKIP DEFENSE & META-GUESSING ENGINE

Recognizing 3-Minute Traps in 15 Seconds, Expected Value Math (+2/-0.5), and Daily Speed Drills.

6.1 Expected Value Engine Proof (+2 / -0.5)

Mathematically, when you eliminate 2 options in a 4-option question, you have a 50% probability of gaining +2.0 marks and a 50% probability of losing -0.5 marks. Expected Value = $(0.5 \times 2) - (0.5 \times 0.5) = +0.75$ marks net gain per question. Over 10 such questions, guessing yields +7.5 net marks! Leaving them blank is a mathematical loss.

Scenario	Probability Matrix	Expected Value (EV)	Action Protocol
Eliminated 2 Options (50/50)	50% win (+2), 50% loss (-0.5)	+0.75 Marks / Q	MANDATORY ATTEMPT
Eliminated 1 Option (1 of 3)	33% win (+2), 67% loss (-0.5)	+0.33 Marks / Q	ATTEMPT
Pure Blind Guess (1 of 4)	25% win (+2), 75% loss (-0.5)	+0.125 Marks	Skip unless total attempts < 65